YEAR 7 · MATHS UNIT 5 · SCIENCE UNIT 4 · HOMEWORK 12

None of the Seven

Part 1: Maths — 5.3 Drawing lines and quadrilaterals, and a review of 5.1 and 5.2
Part 2: Science — 4.1 Characteristics of living organisms and 4.2 Viruses

Name: _____ **Class:** _____

Date given: _____ **Date due:** Monday 16 November 2026

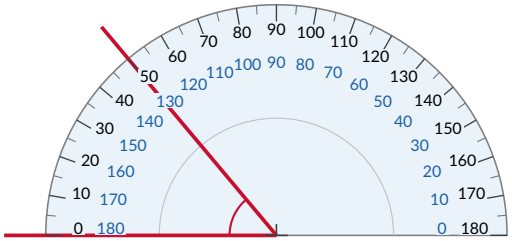
How to do this homework

- **A–B:** maths. **C–D:** science. **No calculator. You do not need a protractor**: a pencil and a ruler are enough.
- Write the **calculation** first, then the answer. **Give a reason** = write the angle fact after **because**.
- Answer in **full sentences** when it says so. *"Respiration"* is not a sentence. *"The gecko gets energy from its food by respiration"* is.

Section A · Maths 5.3 — Lines and Quadrilaterals

A1 Read the protractor

This is a picture of a protractor. You do not need your own.



(a) Look first. Is the red angle **acute** or **obtuse**?

_____

(b) The red line passes **two** numbers. Write them both.

_____ and _____

(c) What size is the red angle? _____

(d) How do you know? Finish the sentence.

"I count up from the 0 that …"

A2 The Workbook's words

Match each question to what it means. Write the **number** in the box. (a) is done for you.

(a) Draw the line through B perpendicular to AB. [4]

(b) Draw a perpendicular line from C to AB. [ ]

(c) Draw a line through P parallel to AB. [ ]

(d) Mark the point D. [ ]

(e) Measure the length of AD. [ ]

(f) How can you check that your answer is correct? [ ]

It means …

1 Use a ruler. Give the length to the nearest mm.

2 Start at C. Stop on AB. Meet AB at 90°.

3 It goes past P. It is always the same distance from AB.

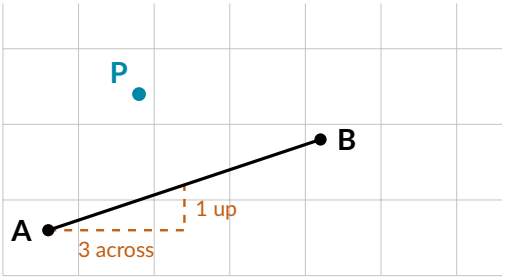
4 It goes past B, at 90° to AB.

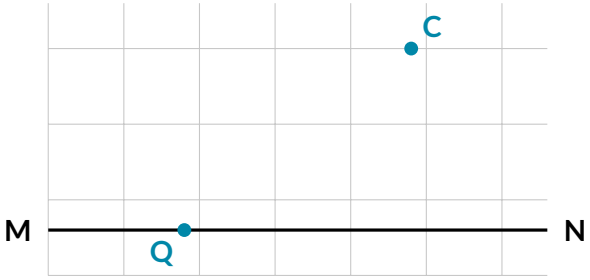
5 Use a fact, like "the four angles add up to 360°".

6 Make a small cross. Write D next to it.

A3 On the grid: parallel and perpendicular

Use a ruler. Count the squares — no protractor.





(a) Draw the line through **P** parallel to **AB**.
AB goes 3 squares across and 1 square up. Your line does the same.

(b) Draw the line through **Q** perpendicular to **MN**.

(c) Draw the perpendicular from **C** to **MN**. Mark the point **D** where it meets MN.

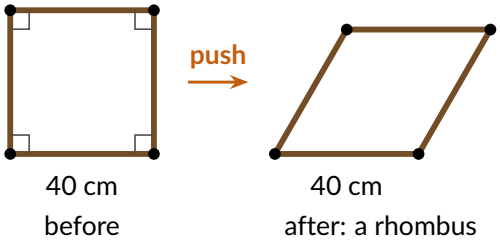
(d) How many squares long is **CD**? _____

(e) Mr Bowen walks from C to N. Is that longer or shorter than CD? Why?

Start: **"CD is the shortest way from C to MN because …"**

A4 A square needs right angles

Mr Bowen makes a square frame from four sticks. Each stick is 40 cm long. He puts **one** nail in each corner. Then he pushes it.



(a) Tick (✓) each box that is true.

Shape	4 equal sides	4 right angles
square		
rhombus		
rectangle		

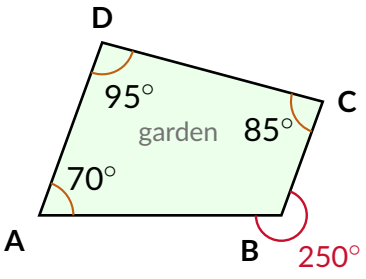
(b) Mr Bowen says: *"My frame still has four sides of 40 cm. So it is still a square."* Is he right? Answer in a full sentence.

Start: **"No, because a square also needs …"**

A5 Does it add up to 360°?

(a) Mr Bowen draws four quadrilaterals. He measures the four angles of each one. Add them up. Is the drawing **accurate**? Row 1 is done for you.

Drawing	The four angles	Add them up	Accurate?
1	85°, 95°, 100°, 80°	85 + 95 + 100 + 80 = 360°	yes
2	70°, 120°, 110°, 50°		yes · no
3	64°, 116°, 64°, 116°		yes · no
4	90°, 90°, 90°, 100°		yes · no



(b) This is Mr Bowen's garden. He measures the angle at **B** and reads **250°**.

(i) Did he measure the inside or the outside?

_____

(ii) What is the angle **inside** the garden at B?

Calculation: _____ **Answer:** _____

(iii) Check: do the four angles inside the garden add up to 360°? Write the calculation.

_____

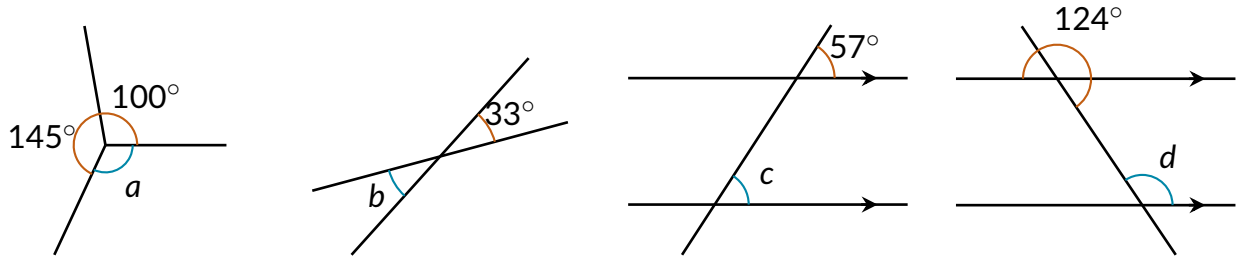
Section B · Review of 5.1 and 5.2 — Calculate, Do Not Measure

B1 Find the angle. Give a reason.

The reasons — copy the one you need after because

line: angles on a straight line add up to 180°
point: angles around a point add up to 360°
shape: angles in a quadrilateral add up to 360°
X: vertically opposite angles are equal
F: corresponding angles are equal
Z: alternate angles are equal

Arrows (>) on two lines mean they are **parallel**. Calculate each letter. Then give a reason.



(a) **(b)** **(c)** **(d)**

a = _____ because _____

b = _____ because _____

c = _____ because _____

d = _____ because _____

Section C · Science 4.1 — Characteristics of Living Organisms

C1 The seven, and one more

Match each word to its meaning. Write the **letter** in the box. The first one is done for you.

A a living thing
B getting bigger
C noticing changes around it
D getting rid of waste, such as carbon dioxide
E changing position or shape
F taking in or making food
G making more organisms of the same kind
H breaking down food inside cells to get energy

organism [A] movement [ ] respiration [ ] sensitivity [ ]

growth [ ] reproduction [ ] excretion [ ] nutrition [ ]

C2 The gecko on the wall

A gecko lives on the wall of Mr Bowen's classroom. Which of the seven is each clue? Row 1 is done for you.

Clue	Which of the seven?
1. The gecko runs up the wall.	movement
2. It sees a fly and turns its head.	
3. It eats the fly.	
4. Inside its cells, the fly is broken down to give energy.	
5. It breathes out carbon dioxide.	
6. It lays two eggs.	
7. A baby gecko is 3 cm long. A month later, it is 5 cm long.	
8. The gecko falls onto Mr Bowen's desk. Mr Bowen sees it and jumps. **Name two** of the seven that Mr Bowen does.	

C3 Breathing or respiration?

Tick (✓) **true** or **false**.

	True	False
1. Respiration is the same as breathing.		
2. Respiration happens inside cells.		
3. Breathing brings in the oxygen that respiration uses.		
4. A tree does not breathe in and out like us, so it does not do respiration.		
5. When you run, your cells do more respiration.		
6. In science, **sensitive** means "gets upset easily".		

Choose **one** false sentence. Write it again so that it is true.

C4 Alive, alive once, or never alive?

Word help

Alive: it is a living organism now. **Alive once:** it came from a living thing, but it is not alive now (a wooden desk was a tree). **Never alive:** it was never a living thing.

Write **A** (alive), **O** (alive once) or **N** (never alive) in each box.

a mango tree ☐ a wooden pencil ☐ a river ☐

a mango seed ☐ a candle flame ☐ a cotton T-shirt ☐

mould on old bread ☐ a glass bottle ☐ a sheet of paper ☐

Mr Bowen says: *"The river is alive. It moves, and it gets bigger in the rainy season."* Is he right? Answer in full sentences.

Start: **"No. A river moves and gets bigger, but it does not …"**

C5 The robot dog and the cactus

Remember from the lesson

A living organism does **all seven**. A non-living thing may do some, but **not all**.

Mr Bowen buys a robot dog. It walks around the room. It barks when someone comes in. When its battery is low, it goes to its charger. It never gets bigger. Mr Bowen says it loves him.

(a) Is the robot dog alive? Answer in full sentences. Name **two** of the seven it seems to do, and **two** it **cannot** do.

Start: **"The robot dog seems to do … But it is not alive because it cannot …"**

(b) A cactus does not walk, and it does not eat. But it **is** alive. Explain, in full sentences. Name **three** of the seven that a cactus does.

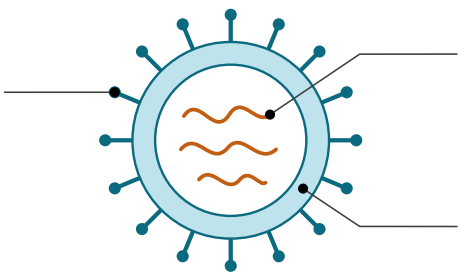
Start: **"A cactus makes its own food using light, so …"**

Section D · Science 4.2 — Viruses

D1 Label the virus

Label each part: **pegs** · **protein coat** · **RNA**. The grey words say what it does.

1 _____
sticks to a cell and helps the virus get inside



2 _____
instructions for making more viruses

3 _____
the outside of the virus

D2 A cell or a virus?

Tick (✓) the **cell**, the **virus**, or **both**.

	A cell in Mr Bowen's nose	A flu virus
1. It has a cell membrane, cytoplasm and a nucleus.		
2. It has a protein coat with pegs.		
3. You can see it with a school microscope.		
4. You can only see it with an electron microscope.		
5. It does respiration.		
6. You cannot see it with your eyes alone.		

D3 Follow the flu virus

Word help

replicate: make copies · **burst**: break open suddenly · **flu**: short for **influenza**

(a) A flu virus replicates. Put the steps in order: write **1** to **5**. Step 1 is done for you.

- [ ] The cell copies the RNA. It builds many new viruses.
- [1] The pegs stick to a cell in the nose.
- [ ] The cell bursts and dies. The new viruses float away.
- [ ] The virus gets inside the cell.
- [ ] The virus bursts open. Its RNA comes out.

(b) Mr Bowen says: *"A virus makes copies of itself, like a photocopier."* What is wrong with this?
Start: **"The virus does not make the copies. The …"**

(c) Why do you feel ill when you have flu?
Start: **"When the new viruses burst out, the cells …"**

D4 Mr Bowen's model virus

Mr Bowen makes a model of a flu virus. He uses a ball of green modelling clay, 20 push pins, and short pieces of red wool, pushed inside the clay.

(a) What is each part of the model?

In the model	In a real virus
the green clay ball	
the push pins	
the red wool	

(b) Give **one** way the model is **not** like a real virus.

Start: **"A real virus is …"**

D5 Is a virus alive?

(a) On its own, outside a cell, does a virus do each of the seven? Write **Y** (yes), **N** (no) or **S** (sort of).

movement ☐ respiration ☐ sensitivity ☐ growth ☐

reproduction ☐ excretion ☐ nutrition ☐

(b) Some scientists say a virus is alive. Others say it is not. What do **you** think? Write 5 sentences.

Start: **"I think a virus is / is not alive because …"**

Your answer must have: tick each one

- ☐ what **you** think, and **because**
- ☐ a reason for the **other** side
- ☐ the word **cells**
- ☐ the word **replicate**
- ☐ one of the **seven**, by name
- ☐ full sentences